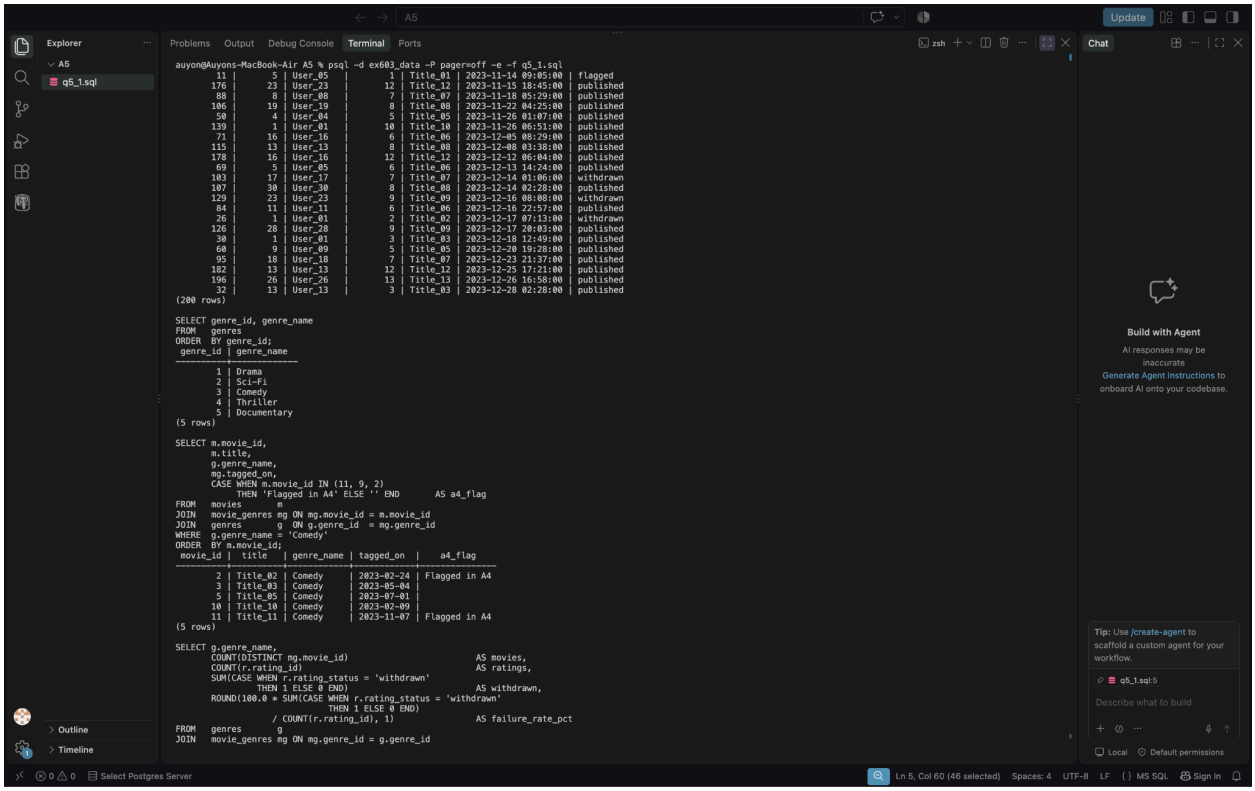
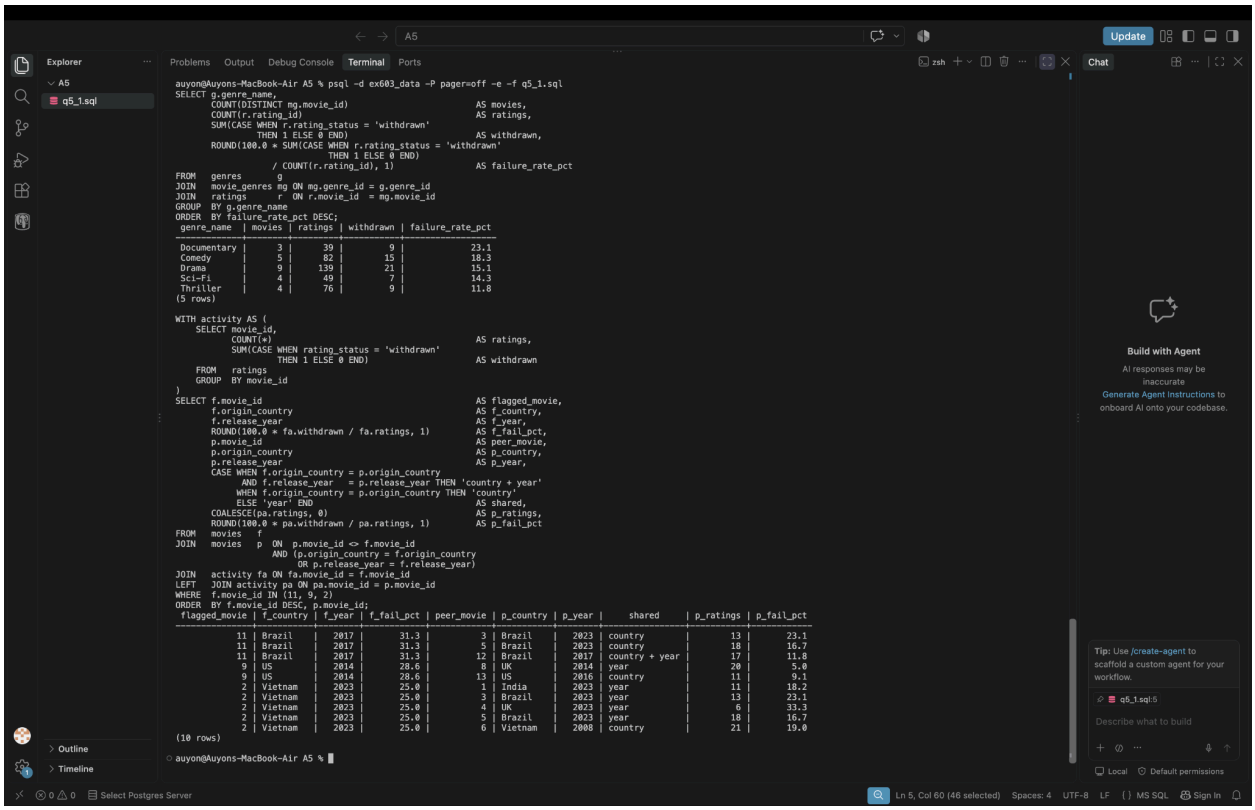


```
auyon@Auyons-MacBook-Air A5 % psql -d ex603_data -P pager=off -e -f q5_1.sql
        11 |       5 | User_05   |       1 | Title_01 | 2023-11-14 09:05:00 | flagged
       176 |      23 | User_23   |      12 | Title_12 | 2023-11-15 18:45:00 | published
        88 |       8 | User_08   |       7 | Title_07 | 2023-11-18 05:29:00 | published
       106 |      19 | User_19   |       8 | Title_08 | 2023-11-22 04:25:00 | published
        50 |       4 | User_04   |       5 | Title_05 | 2023-11-26 01:07:00 | published
       139 |       1 | User_01   |      10 | Title_10 | 2023-11-26 06:51:00 | published
        71 |      16 | User_16   |       6 | Title_06 | 2023-12-05 08:29:00 | published
       115 |      13 | User_13   |       8 | Title_08 | 2023-12-08 03:38:00 | published
       178 |      16 | User_16   |      12 | Title_12 | 2023-12-12 06:04:00 | published
        69 |       5 | User_05   |       6 | Title_06 | 2023-12-13 14:24:00 | published
       103 |      17 | User_17   |       7 | Title_07 | 2023-12-14 01:06:00 | withdrawn
       107 |      30 | User_30   |       8 | Title_08 | 2023-12-14 02:28:00 | published
       129 |      23 | User_23   |       9 | Title_09 | 2023-12-16 08:08:00 | withdrawn
        84 |      11 | User_11   |       6 | Title_06 | 2023-12-16 22:57:00 | published
        26 |       1 | User_01   |       2 | Title_02 | 2023-12-17 07:13:00 | withdrawn
       126 |      28 | User_28   |       9 | Title_09 | 2023-12-17 20:03:00 | published
        30 |       1 | User_01   |       3 | Title_03 | 2023-12-18 12:49:00 | published
        60 |       9 | User_09   |       5 | Title_05 | 2023-12-20 19:28:00 | published
        95 |      18 | User_18   |       7 | Title_07 | 2023-12-23 21:37:00 | published
       182 |      13 | User_13   |      12 | Title_12 | 2023-12-25 17:21:00 | published
       196 |      26 | User_26   |      13 | Title_13 | 2023-12-26 16:58:00 | published
        32 |      13 | User_13   |       3 | Title_03 | 2023-12-28 02:28:00 | published
(200 rows)

SELECT genre_id, genre_name
FROM   genres
ORDER  BY genre_id;
 genre_id | genre_name
----------+-------------
        1 | Drama
        2 | Sci-Fi
        3 | Comedy
        4 | Thriller
        5 | Documentary
(5 rows)

SELECT m.movie_id,
       m.title,
       g.genre_name,
       mg.tagged_on,
       CASE WHEN m.movie_id IN (11, 9, 2)
            THEN 'Flagged in A4' ELSE '' END       AS a4_flag
FROM   movies       m
JOIN   movie_genres mg ON mg.movie_id = m.movie_id
JOIN   genres       g  ON g.genre_id  = mg.genre_id
WHERE  g.genre_name = 'Comedy'
ORDER  BY m.movie_id;
 movie_id |  title   | genre_name | tagged_on  |    a4_flag
----------+----------+------------+------------+---------------
        2 | Title_02 | Comedy     | 2023-02-24 | Flagged in A4
        3 | Title_03 | Comedy     | 2023-05-04 |
        5 | Title_05 | Comedy     | 2023-07-01 |
       10 | Title_10 | Comedy     | 2023-02-09 |
       11 | Title_11 | Comedy     | 2023-11-07 | Flagged in A4
(5 rows)

SELECT g.genre_name,
       COUNT(DISTINCT mg.movie_id)                          AS movies,
       COUNT(r.rating_id)                                   AS ratings,
       SUM(CASE WHEN r.rating_status = 'withdrawn'
                THEN 1 ELSE 0 END)                          AS withdrawn,
       ROUND(100.0 * SUM(CASE WHEN r.rating_status = 'withdrawn'
                              THEN 1 ELSE 0 END)
                   / COUNT(r.rating_id), 1)                 AS failure_rate_pct
FROM   genres       g
JOIN   movie_genres mg ON mg.genre_id = g.genre_id
```



```
auyon@Auyons-MacBook-Air A5 % psql -d ex603_data -P pager=off -e -f q5_1.sql
SELECT g.genre_name,
       COUNT(DISTINCT mg.movie_id)                          AS movies,
       COUNT(r.rating_id)                                   AS ratings,
       SUM(CASE WHEN r.rating_status = 'withdrawn'
                THEN 1 ELSE 0 END)                          AS withdrawn,
       ROUND(100.0 * SUM(CASE WHEN r.rating_status = 'withdrawn'
                              THEN 1 ELSE 0 END)
                   / COUNT(r.rating_id), 1)                 AS failure_rate_pct
FROM   genres       g
JOIN   movie_genres mg ON mg.genre_id = g.genre_id
JOIN   ratings      r  ON r.movie_id  = mg.movie_id
GROUP  BY g.genre_name
ORDER  BY failure_rate_pct DESC;
 genre_name  | movies | ratings | withdrawn | failure_rate_pct
-------------+--------+---------+-----------+------------------
 Documentary |      3 |      39 |         9 |             23.1
 Comedy      |      5 |      82 |        15 |             18.3
 Drama       |      9 |     139 |        21 |             15.1
 Sci-Fi      |      4 |      49 |         7 |             14.3
 Thriller    |      4 |      76 |         9 |             11.8
(5 rows)

WITH activity AS (
    SELECT movie_id,
           COUNT(*)                                         AS ratings,
           SUM(CASE WHEN rating_status = 'withdrawn'
                    THEN 1 ELSE 0 END)                      AS withdrawn
    FROM   ratings
    GROUP  BY movie_id
)
SELECT f.movie_id                                           AS flagged_movie,
       f.origin_country                                     AS f_country,
       f.release_year                                       AS f_year,
       ROUND(100.0 * fa.withdrawn / fa.ratings, 1)          AS f_fail_pct,
       p.movie_id                                           AS peer_movie,
       p.origin_country                                     AS p_country,
       p.release_year                                       AS p_year,
       CASE WHEN f.origin_country = p.origin_country
             AND f.release_year   = p.release_year THEN 'country + year'
            WHEN f.origin_country = p.origin_country THEN 'country'
            ELSE 'year' END                                 AS shared,
       COALESCE(pa.ratings, 0)                              AS p_ratings,
       ROUND(100.0 * pa.withdrawn / pa.ratings, 1)          AS p_fail_pct
FROM   movies   f
JOIN   movies   p  ON  p.movie_id <> f.movie_id
                   AND (p.origin_country = f.origin_country
                        OR p.release_year = f.release_year)
JOIN   activity fa ON fa.movie_id = f.movie_id
LEFT   JOIN activity pa ON pa.movie_id = p.movie_id
WHERE  f.movie_id IN (11, 9, 2)
ORDER  BY f.movie_id DESC, p.movie_id;
 flagged_movie | f_country | f_year | f_fail_pct | peer_movie | p_country | p_year |     shared     | p_ratings | p_fail_pct
---------------+-----------+--------+------------+------------+-----------+--------+----------------+-----------+------------
            11 | Brazil    |   2017 |       31.3 |          3 | Brazil    |   2023 | country        |        13 |       23.1
            11 | Brazil    |   2017 |       31.3 |          5 | Brazil    |   2023 | country        |        18 |       16.7
            11 | Brazil    |   2017 |       31.3 |         12 | Brazil    |   2017 | country + year |        17 |       11.8
             9 | US        |   2014 |       28.6 |          8 | UK        |   2014 | year           |        20 |        5.0
             9 | US        |   2014 |       28.6 |         13 | US        |   2016 | country        |        11 |        9.1
             2 | Vietnam   |   2023 |       25.0 |          1 | India     |   2023 | year           |        11 |       18.2
             2 | Vietnam   |   2023 |       25.0 |          3 | Brazil    |   2023 | year           |        13 |       23.1
             2 | Vietnam   |   2023 |       25.0 |          4 | UK        |   2023 | year           |         6 |       33.3
             2 | Vietnam   |   2023 |       25.0 |          5 | Brazil    |   2023 | year           |        18 |       16.7
             2 | Vietnam   |   2023 |       25.0 |          6 | Vietnam   |   2008 | country        |        21 |       19.0
(10 rows)

auyon@Auyons-MacBook-Air A5 %
```